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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Seventh Semester B.Tech Degree (S, FE) Examination May 2023 (2019 Scheme)

Course Code: CST413**Course Name: MACHINE LEARNING****Max. Marks: 100****Duration: 3 Hours****PART A***Answer all questions, each carries 3 marks.*

Marks

- 1 Differentiate between Maximum Likelihood estimation (MLE) and Maximum a Posteriori (MAP) estimation? (3)
- 2 Find the ML estimate of the parameter p in the geometric distribution defined by the probability mass function $f(x) = (1-p)p^x, x=1,2,3,\dots$ (3)
- 3 Specify the basic principle of gradient descent algorithm (3)
- 4 Find a quadratic regression model for the following data (3)

x	3	4	5	6	7
y	2.5	3.2	3.8	6.5	11.5

- 5 Why do you use kernels to model a projection from attributes into a feature space, instead of simply projecting the dataset directly? (3)
- 6 ReLU activation functions are most used in neural networks instead of the tanh activation function. Draw both activation functions and give
 - a) an advantage of the ReLU function compared to the tanh function.
 - b) a disadvantage of the ReLU function compared to the tanh function.
 (3)
- 7 Find similarity measure to compare feature vectors of two images? Justify your answer. (3)
- 8 Describe agglomerative hierarchical clustering? (3)
- 9 Classifier A attains 100% accuracy on the training set and 70% accuracy on the test set. Classifier B attains 70% accuracy on the training set and 75% accuracy on the test set. Which one is a better classifier. Justify your answer. (3)
- 10 State Area under curve in machine learning? Specify the significance of AUC (3)

PART B*Answer any one full question from each module, each carries 14 marks.***Module I**

- 11 a) Distinguish between classification and regression with an example. (4)

- b) Suppose that X is a discrete random variable with the following probability mass function where $0 \leq \theta \leq 1$ is a parameter. The following 10 independent observations were taken from such a distribution: (3, 0, 2, 1, 3, 2, 1, 0, 2, 1). What is the MLE of θ . (10)

X	0	1	2	3
P(x)	$2\theta/3$	$\theta/3$	$2(1-\theta)/3$	$(1-\theta)/3$

OR

- 12 a) Describe in detail with examples (8)
 i)Supervised learning ii)Unsupervised learning iii)Reinforcement learning
- b) Suppose you have a three class problem where class label $y \in \{0, 1, 2\}$ and each training example X has 3 binary attributes $X_1, X_2, X_3 \in \{0, 1\}$. How many parameters (probability distribution) do you need to know to classify an example using the Naive Bayes classifier? (6)

Module II

- 13 a) Identify the first splitting attribute for decision tree by using ID3 algorithm with the following dataset. (10)

Age	Competition	Type	Class(Profit)
Old	Yes	Software	Down
Old	No	Software	Down
Old	No	Hardware	Down
Mild	Yes	Software	Down
Mild	Yes	Hardware	Down
Mild	No	Hardware	Up
Mild	No	Software	Up
New	Yes	Software	Up
New	No	Hardware	Up
New	No	Software	Up

- b) Given the set of values $X = (3, 9, 11, 5, 2)^T$ and $Y = (1, 8, 11, 4, 3)^T$. Evaluate the regression coefficients. (4)

OR

- 14 a) Given the following data on a certain set of patients seen by a doctor, can the doctor conclude that a person having chills, fever, and mild headache and without running nose has the flu? (Use Bayesian Classification) (7)

Chills	Running nose	Headache	Fever	Has Flu
Y	N	Mild	Y	N
Y	Y	No	N	Y
Y	N	Strong	Y	Y
N	Y	Mild	Y	Y
N	N	No	N	N
N	Y	Strong	Y	Y
N	Y	Strong	N	N
Y	Y	Mild	Y	Y

- b) Consider the hypothesis for the linear regression $h_0(x) = \Theta_0 + \Theta_1 x$, and the cost function $J(\Theta_0, \Theta_1) = 1/2m \sum_{i=1}^m (h_0(x^{(i)}) - y^{(i)})^2$ where m is the number of training examples. Given the following set of training examples. (7)

x	y
3	2
1	2
0	1
4	3

Answer the following questions :

- 1) Find the value of $h_0(2)$ if $\Theta_0 = 0$ and $\Theta_1 = 1.5$
- 2) Find the value of $J(0,1)$
- 3) Suppose the value of $J(\Theta_0, \Theta_1) = 0$. What can be inferred from this?

Module III

- 15 a) Given the following data, use ANN with one hidden layer, appropriate initial weights and biases to compute the optimal values of the weights. Perform one iteration of the forward and phases of the back propagation algorithm for each sample. (10)

Sample	Input1	Input2	Output1	Output2
1	1.20	2.30	0.53	0.76
2	0.23	0.37	1.17	2.09

- b) Calculate the output y of a three input neuron with bias. The input feature vector is $(x_1, x_2, x_3) = (0.8, 0.6, 0.4)$ and weight values are $[w_1, w_2, w_3, b] = [0.2, 0.1, -0.3, 0.35]$. Use binary Sigmoid function as activation function. (4)

OR

- 16 a) State the mathematical formulation of the SVM problem. Give an outline of the method for solving the problem (8)
- b) Show the basic idea of back propagation algorithm (6)

Module IV

- 17 a) Suppose that we have the following data: (2, 0), (1, 2), (2, 2), (3, 2), (2, 3), (3, 3), (2, 4), (3, 4), (4, 4), (3, 5) Identify the cluster by applying the k-means algorithm, with $k = 2$. Try using initial cluster centres as far apart as possible (8)
- b) Illustrate the idea of PCA for a two dimensional data using suitable diagrams (6)

OR

- 18 a) Given the following data, use PCA to reduce the dimension (10)

Feature	Example1	Example2	Example3
X	2	3	7
y	11	14	26

- b) Given two objects represented by the tuples (22, 1, 42, 10) and (20, 0, 36, 8): (4)
- (i) Compute the Euclidean distance between the two objects.
- (ii) Compute the Manhattan distance between the two objects.

Module V

- 19 a) Suppose there are three classifiers A, B and C. The (FPR, TPR) measures of the three classifiers are as follows – A (0, 1), B (1, 1), C (1, 0.5). Which can be considered as a perfect classifier? Justify your answer. (6)
- b) Suppose 10000 patients get tested for flu; out of them, 9000 are actually healthy and 1000 are actually sick. For the sick people, a test was positive for 620 and negative for 380. For the healthy people, the same test was positive for 180 and negative for 8820. Construct a confusion matrix for the data and compute the accuracy, precision and recall for the data. Also construct the ROC curve. (8)

OR

- 20 a) Assume you have a model with a high bias and a low variance. Interpret the characteristics of such a model? (6)
- b) State ROC space and ROC curve in machine learning? In ROC space, which points correspond to perfect prediction, always positive prediction and always negative prediction? Why? (8)
